Calculus of Variation

Assignment 6

Q1 Find the curves for which the functional

$$I = \int_0^{x_1} \frac{\sqrt{1+y'^2}}{y} dx$$

with y(0) = 0 can have extrema if the point (x_1, y_1) can vary along the line y = x - 5

a) $y = \pm\sqrt{x - 2x^2}$

b) $y = \pm\sqrt{2x + 3x^2}$

c) $y = \pm\sqrt{x - 5x^2}$

d) $y = \pm\sqrt{3x - x^2}$

e) $y = \pm\sqrt{10x + 5x^2}$

f) $y = \pm\sqrt{10x - x^2}$

Q2 Find the curves for which the functional

$$I = \int_0^{x_1} \frac{\sqrt{1+y'^2}}{y} dx$$

with y(0) = 0 can have extrema if the point (x_1, y_1) can vary along the circle $(x-9)^2 + y^2 = 9$

a) $y^2 = x^2 + 4x$

b) $y^2 = -x^2 + 3x - 1$

c) $y^2 = -x^2 + 5x + 3$

d) $y^2 = -3x^2 + 2x + 1$

e) $y^2 = -x^2 + 8x$

f) $y^2 = x^2 - x$

Q3 Consider general functionals of the form

$$I = \int_0^{x_1} k(x,y) e^{\tan^{-1} y'} \sqrt{1+y'^2} dx$$

where k(x, y) $\neq$ 0 and the desired extremals are subject to the endpoint conditions that (x_o, y_o) is prescribed and fixed but (x_1, y_1) is only constrained to lie on the curve $y = \phi(x)$. Find the traversality condition that applies at (x_1, y_1)

a) $(1-y', 1+y')(0, \phi')$

b) $(1+y', 1-y')(0, \phi')$

c) $(1+y', 1-y')(1, \phi')$

d) $(1-y', 1+y')(1, \phi')$

e) $(1-y', 1+y')(\phi', 1)$

f) $(1+y', 1-y')(\phi', 1)$